Git

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What is Git?
Version control system (DVCS) used to track changes in source code and manage software development projects.

It was created by Linus Trovalds in 2005 for the development of the Linux kernel.

Simple Definition
Git is a tool that helps developers:

- Track changes in file
- Maintain different versions of a project
- Collaborate with other developers
- Restore previous versions when needed

Why do we need git?
Imagine you are developing a project.

Without git:

```
Project_final.py
Project_Final_new.py
Project_Final_Latest.py
Project_Final_Really_Final.py
```

Managing versions become difficult.

```
Project=>
        |=>Git Repository
           |=> version 1
           |=> version 2
           |=> version 3
           |=> version 4
```

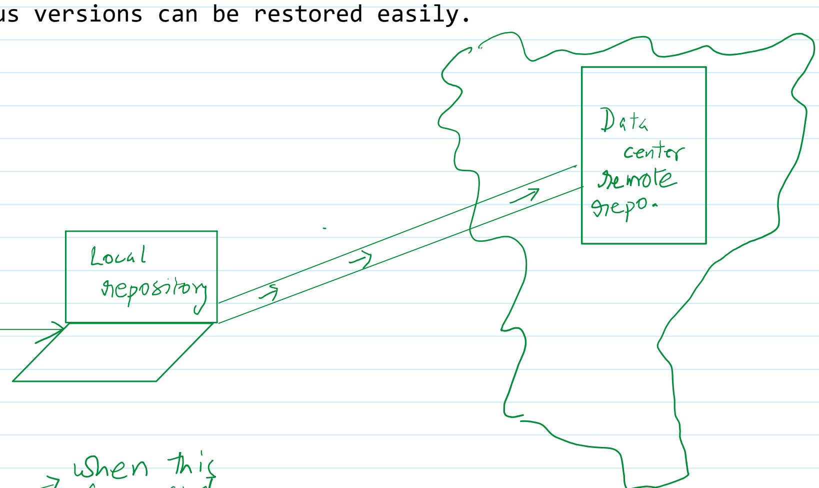
Git automatically tracks every change.

Features of Git

1. **Version control:** Tracks every modification made to files.
2. **Distributed System:** Every developer has a complete copy of the repository.
3. **Fast performance:** Operations are performed locally.
4. **Branching:** Allows development of new features without affecting the main project.
5. **Collaboration:** Multiple developers can work on the same project simultaneously.
6. **Backup and Recovery:** Previous versions can be restored easily.

Git Architecture

```
Working Directory
      |
      V
   git add
      |
      V
  Staging area
      |
      V
  git commit
      |
      V
Local repository
      |
      V
   git push
      |
      V
Remote repository
```



when this command is issued, it transfer local repository snapshot to remote repository and creates a new version (commit) in git.

Git terminology
Repository(Repo)
A repository is a storage location for your project.

Project files + git History = Repository

Commit: A commit is a snapshot of your project at a specific point in time.

Branch: A branch is an independent line of development.

```
   Main
     |
     |--feature-login
     |--feature-payment
```

Merge: Combines changes from different branches.

```
       feature-login
         |
         V
       merge
         |
         V
       main
```

WAP to input a number and count number of digits present in it using recursion:

```
def countDigit(num):
   pass

#main code
N = int(input('Enter any positive integer value:'))
C = countDigit(N)
print(f'Number of digits is: {C}')
```

Output:
Enter any positive integer value: 346
Number of digits is: 3

main code

num = 99

```
# WAP to input a number and count number of digits present in it using recursion:

def countDigit(num):
    if num == 0:
        return 0
    else:
        return 1+countDigit(num//10)

#main code
N = int(input('Enter any positive integer value:'))
C = countDigit(N)
print(f'Number of digits is: {C}')

# Output:
# Enter any positive integer value: 346
# Number of digits is: 3
```

99 9 0

due: 1 = 2
1 + countDigit(9)

num = 9 0 = 1
1 + countDigit(0)

num = 0
return 0

Wap to input number of terms to print and find the sum of following series using recursion:

Number of terms : 3

(a) 1 + 4 + 9 + ...

(b) $1 + \frac{1}{2} + \frac{1}{3} +$ ....

(c) $1 + \frac{1}{3} + \frac{1}{5} +$ ....

(d) $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} +$ ....

(e) 2 - 4 + 6 - 8 + ...

(f) (1*2) + (2*3) + ...

$\boxed{1+4+9 = 14}$ output

output

$\boxed{1 + 1/2 + 1/3 = 1.08}$

$1 + 0.5 + 0.\bar{3} = 1.08$

Number of terms = 5

$1 + 4 + 9 + 16 + 25 = 55$

```
def sumOfSeries(terms):
    if terms > 1
        return terms**2 + sumOfSeries(terms-1)
        print(terms, '+')
    else:
        return terms**2
        print(terms)
```

```
def sumOfSeries(terms):
    if terms > 1:
        print(terms**2, ' + ', end='')
        return terms**2 + sumOfSeries(terms-1)

    else:
        print(terms**2, ' = ', end = '')
        return terms**2


#main code
N = int(input('Enter number of terms: '))
sum = sumOfSeries(N)
print(sum)
```

main code

terms = 5   30 = 55
return 5^2 + ~~sumOfSeries(4)~~

terms = 4   14 = 30
return 4^2 + ~~sumOfSeries(3)~~

terms = 3   5 = 14
return 3^2 + ~~sumOfSeries(2)~~

terms = 2   1 = 5
return 2^2 + ~~sumOfSeries(1)~~

return 1^2

```
Output:
Enter number of terms: 5
25  + 16  + 9  + 4  + 1  = 55
```

sum from main code.

```
def sumOfSeries(terms, start = 1):
    if start < terms:
        print(start**2, ' + ', end='')
        return start**2 + sumOfSeries(terms, start+1)
    else:
        print(start**2, ' = ', end = '')
        return start**2

#main code
N = int(input('Enter number of terms: '))
sum = sumOfSeries(N)
print(sum)
```

Debugging of code is very much important skill.